

Seventh Biennial Ptolemy Miniconference Program

February 13, 2007

Registration and Overview

- 8:00 - 8:30 Registration and Continental Breakfast
HP Auditorium, [3rd Floor Soda Hall, UC Berkeley](#)
- 8:30 - 8:45 **"Ptolemy Project Overview, Status, and Future Directions"**
Edward A. Lee, UC Berkeley
- 8:45 - 9:15 **"Kepler Project Overview, Status, and Future Directions"**
Matthew Jones, National Center for Ecological Analysis and Synthesis (NCEAS)

Session 1

- 9:15 - 9:35 **"The Kepler Actor Repository: Enabling Remote Storage, Query and Retrieval of Actors and other Workflow Components"**
Chad Berkley, NCEAS
- 9:35 - 9:55 **"Leveraging synchronous language principles for continuous-time, discrete-event, and hybrid system models."**
Haiyang Zheng, UC Berkeley
- 9:55 - 10:15 Break

Session 2

- 10:15 - 10:35 **"Scheduling Data-Intensive Workflows in Kepler"**
Tim H. Wong
- 10:35 - 10:55 **"PTIDES: Programming Temporally Integrated Distributed Embedded Systems"**
Yang Zhao, UC Berkeley
- 10:55 - 11:15 **"Flexible Scientific Workflow with Dynamic Embedding"**
Anne Hee Hiong Ngu, Texas State
- 11:15 - 11:35 **"Partial Evaluation for Optimized Compilation of Actor-oriented Models"**
Gang Zhou

Posters and Lunch

- 11:35 - 12:00 Poster Overviews Poster presenters are encouraged to give a 3 minute presentation about their poster. ([instructions](#))
- "Real-Time Fault-Tolerant Distributed DE for Embedded Systems"
Thomas Feng, UC Berkeley
- "A Kepler-based Three Tier Architecture applied to LiDAR Interpolation"
Efrat Frank, Ilkay Altintas, San Diego Supercomputer Center
- "Nesting and Unnesting Models of Computation"
Antoon Goderisa, University of Manchester
- "Using the RExpression Actor in Kepler Workflows"
Dan Higgins & Josh Madin, NCEAS
- "C code generation framework in Ptolemy II"
Man-Kit Leung, UC Berkeley

"Actor Oriented as OpenOffice.org Macro Language"
 Rodolfo Totaro, SP-Process s.p.a.
 "Ptolemy Reconfigurable and Programmable Hardware Platform"
 Mohamed A. Salem, Mentor Graphics Corporation
 "How to measure Multi-Instruction, Multi-Core Processor
 Performance using Simulation"
 Deepak Shankar & Darryl Koivisto, Mirabilis Design Inc.
 "Overcoming the Gap Between Design at Electronic System Level
 (ESL) and Implementation"
 Horst Salzwedel, Technical University of Ilmenau

12:00 - Lunch (Provided)
 12:30 Wozniak Lounge, 4th floor Soda Hall
 12:30 - 2:00 Poster Session
 Wozniak Lounge

Session 3

2:00 - 2:20 **"Correctness in Causal Systems"**
 Eleftherios Matsikoudis, UC Berkeley
 2:20 - 2:40 **"Causality Interfaces for Actor Networks"**
 Ye Zhou, UC Berkeley
 2:40 - 3:00 **"COMAD:Collection-Oriented Modeling and Design of Scientific
 Workflows and Data"**
 Timothy McPhillips, UC Davis
 3:00 - 3:20 **"The Power of Higher-Order Components in System Design"**
 Adam Cataldo, UC Berkeley
 3:20 - 3:40 Break

Session 4

3:40 - 4:00 Tentative topic: Provenance
 Tentative speakers: Ilkay Altintas & Norbert Podhorszki
 4:00 - 4:20 **"Viptos: A Graphical Development and Simulation Environment for
 TinyOS-based Wireless Sensor Networks"**
 Elaine Cheong, UC Berkeley
 4:20 - 4:40 Tentative title: **"The CAL Actor Language: Synthesizing Models to
 FPGA"**
 Jörn Janneck, Xilinx Research Labs
 4:40 - 5:00 "Cyber-Physical Systems: A Vision of the Future"
 Edward A. Lee, UC Berkeley
 5:00 - 6:30 Reception
 Wozniak Lounge